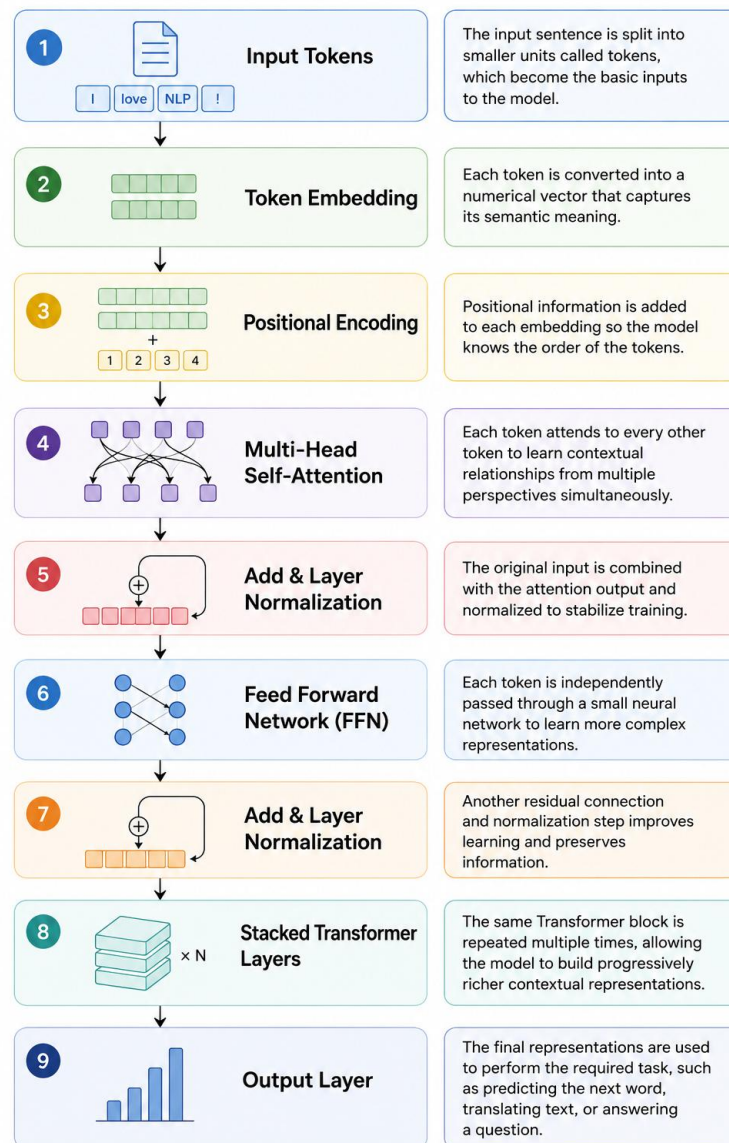




Transformer Architecture

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Why Do We Need Transformers?

Consider the sentence:

The book that I borrowed yesterday from the library was fascinating because it explained quantum mechanics beautifully.

To understand what "it" refers to, the model needs to remember information from many words earlier. As sentences become longer, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) struggle to retain this context effectively.

Transformers were introduced to overcome this limitation by processing all words simultaneously instead of sequentially. This allows them to better understand relationships between words regardless of how far apart they appear in a sentence.

What is a Transformer?

A Transformer is a deep learning architecture introduced in the paper "**Attention Is All You Need**" (2017).

Instead of reading a sentence word by word, a Transformer analyses the entire sentence at once. It uses a mechanism called **Self-Attention** to determine which words are important while understanding the meaning of every other word.

It's like reading a paragraph, we don't memorize every word in order; instead, we quickly identify the important words and relate them to one another. Transformers attempt to learn language in a similar way.

How Does a Transformer Work?

1. Input Tokens

The input sentence is split into smaller units called tokens, which become the basic inputs to the model.

2. Token Embedding

Each token is converted into a numerical vector that captures its semantic meaning.

3. Positional Encoding

Positional information is added to each embedding so the model knows the order of the tokens.

4. Multi-Head Self-Attention

Each token attends to every other token to learn contextual relationships from multiple perspectives simultaneously.

5. Add & Layer Normalization

The original input is combined with the attention output and normalized to stabilize training.

6. Feed Forward Network (FFN)

Each token is independently passed through a small neural network to learn more complex representations.

7. Add & Layer Normalization

Another residual connection and normalization step improves learning and preserves information.

8. Stacked Transformer Layers

The same Transformer block is repeated multiple times, allowing the model to build progressively richer contextual representations.

9. Output Layer

The final representations are used to perform the required task, such as predicting the next word, translating text, or answering a question.

Key Takeaways

- Transformers process an entire sentence simultaneously instead of one word at a time.
- Self-Attention allows every word to understand its relationship with every other word.
- Parallel processing makes Transformers significantly faster than RNNs and LSTMs.
- Modern language models such as GPT, BERT, and T5 are all built using the Transformer architecture.